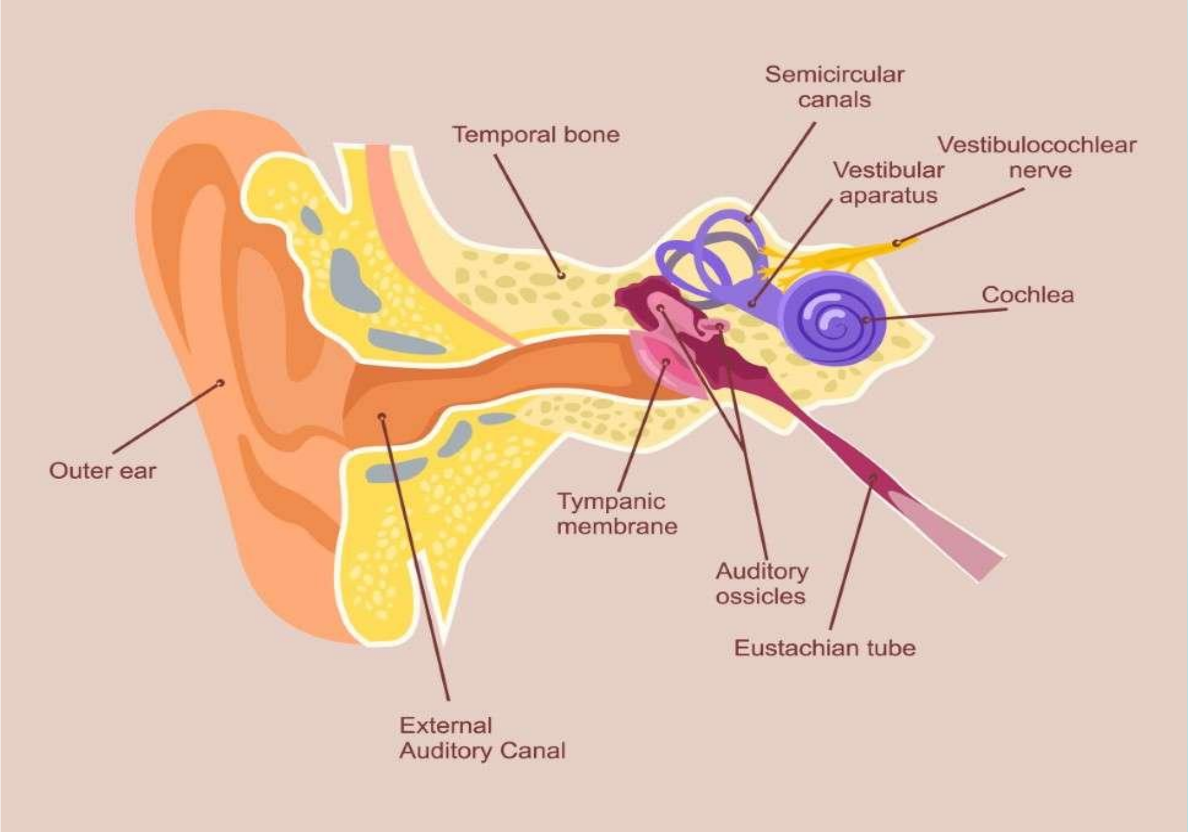
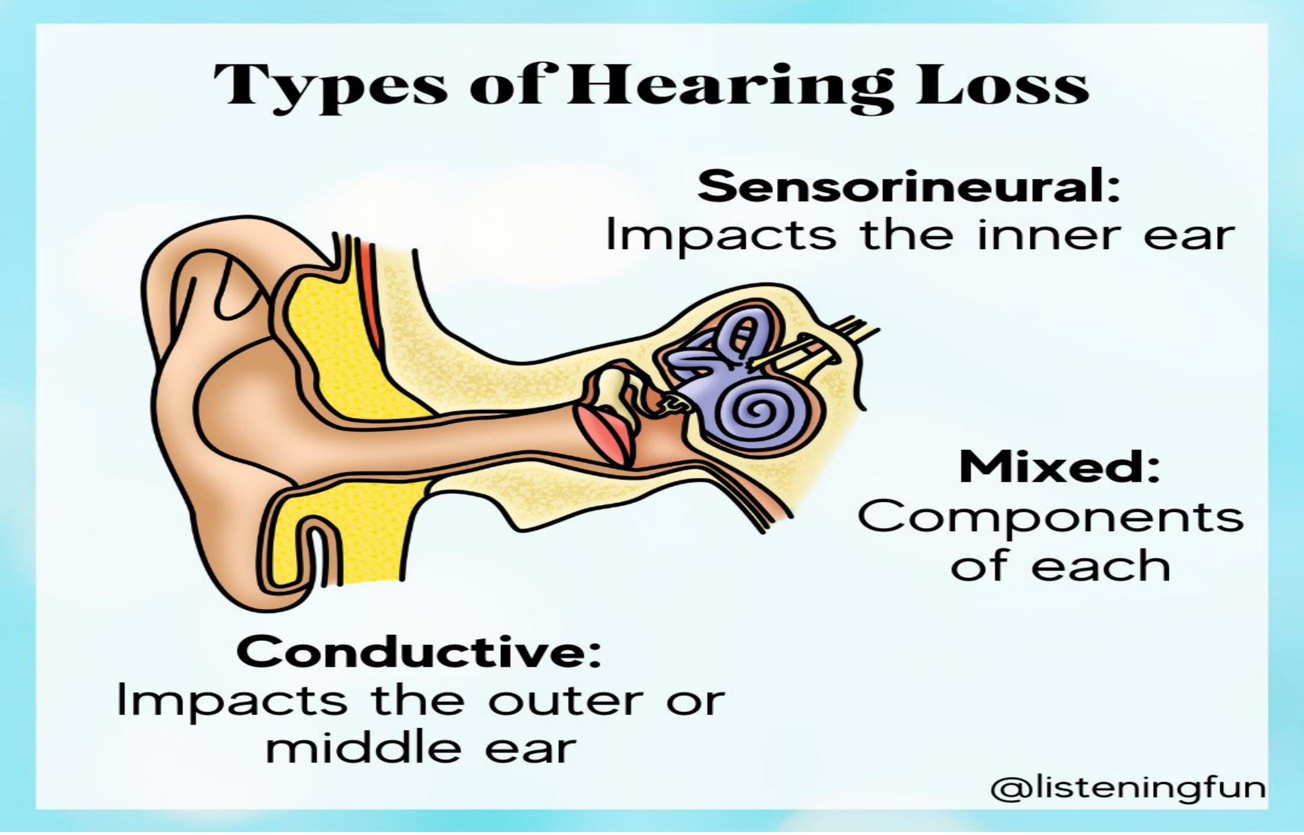
Classify types of hearing loss? Explain causes of hearing loss, relate types of hearing loss to clinical features





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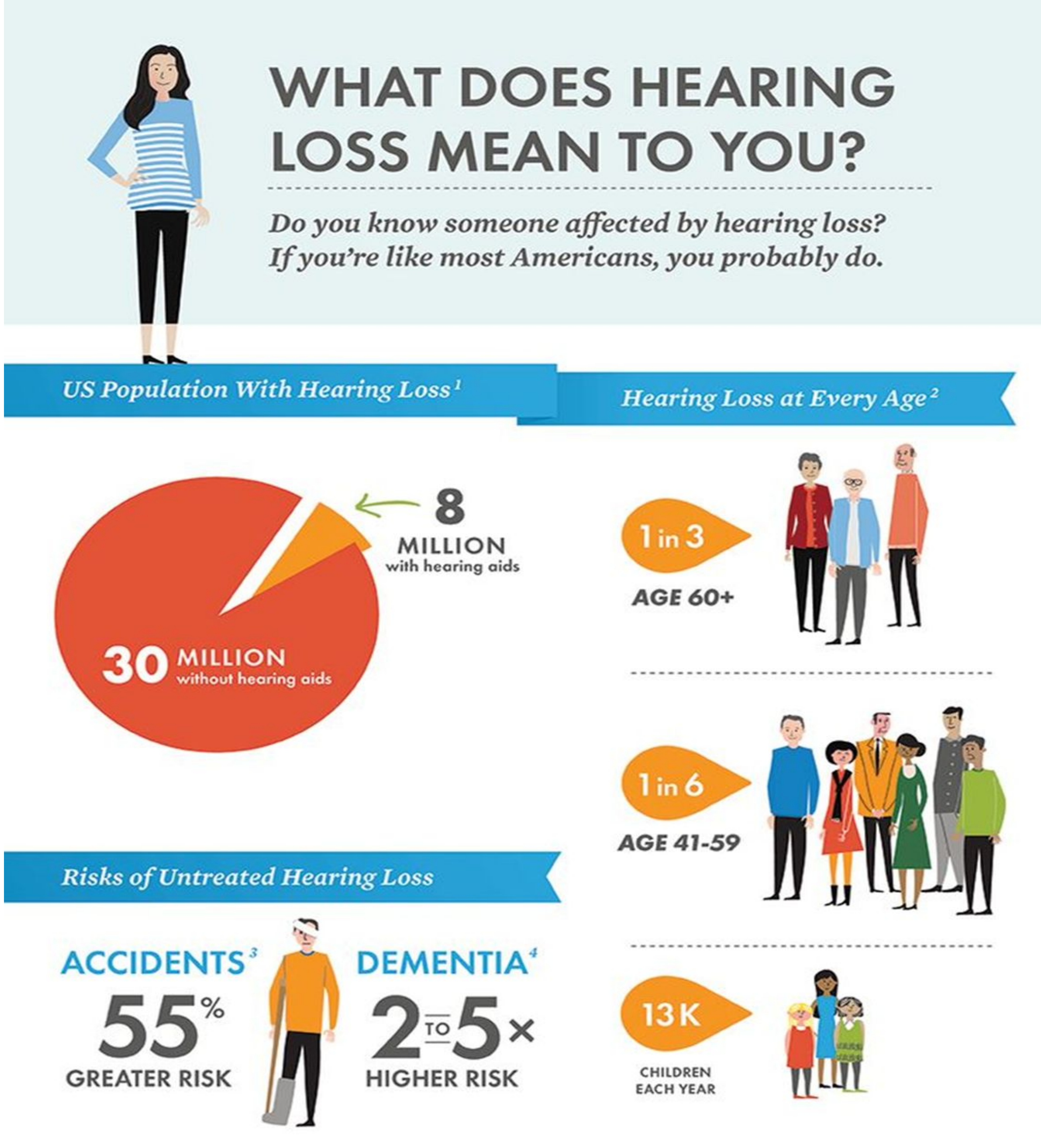
Definition

Hearing impairment is when an individual loses the ability to hear in either one or both ears.

The level of impairment can vary from mild to severe or total loss of hearing.

Facts

- Nearly 300 million people worldwide suffer from some hearing loss in both ears.
- The majority of those people live in developing or middle class countries.
    - About 35 million Americans report some hearing loss.
- 2 to 3 out of 1000 children born in the United States are born deaf or hard-of-hearing.



Hearing Loss Worldwide: By the Numbers

466 million
Number of people now experiencing hearing loss

900 million
Number of people estimated to have hearing loss by 2050

60%
Percentage of childhood hearing loss resulting from preventable causes

1.1 billion
Number of people between 12 and 35 years old at risk of hearing loss due to noise exposure

$750 billion
Annual global cost of untreated hearing loss, in U.S. dollars

Source: World Health Organization

Types of Hearing Loss

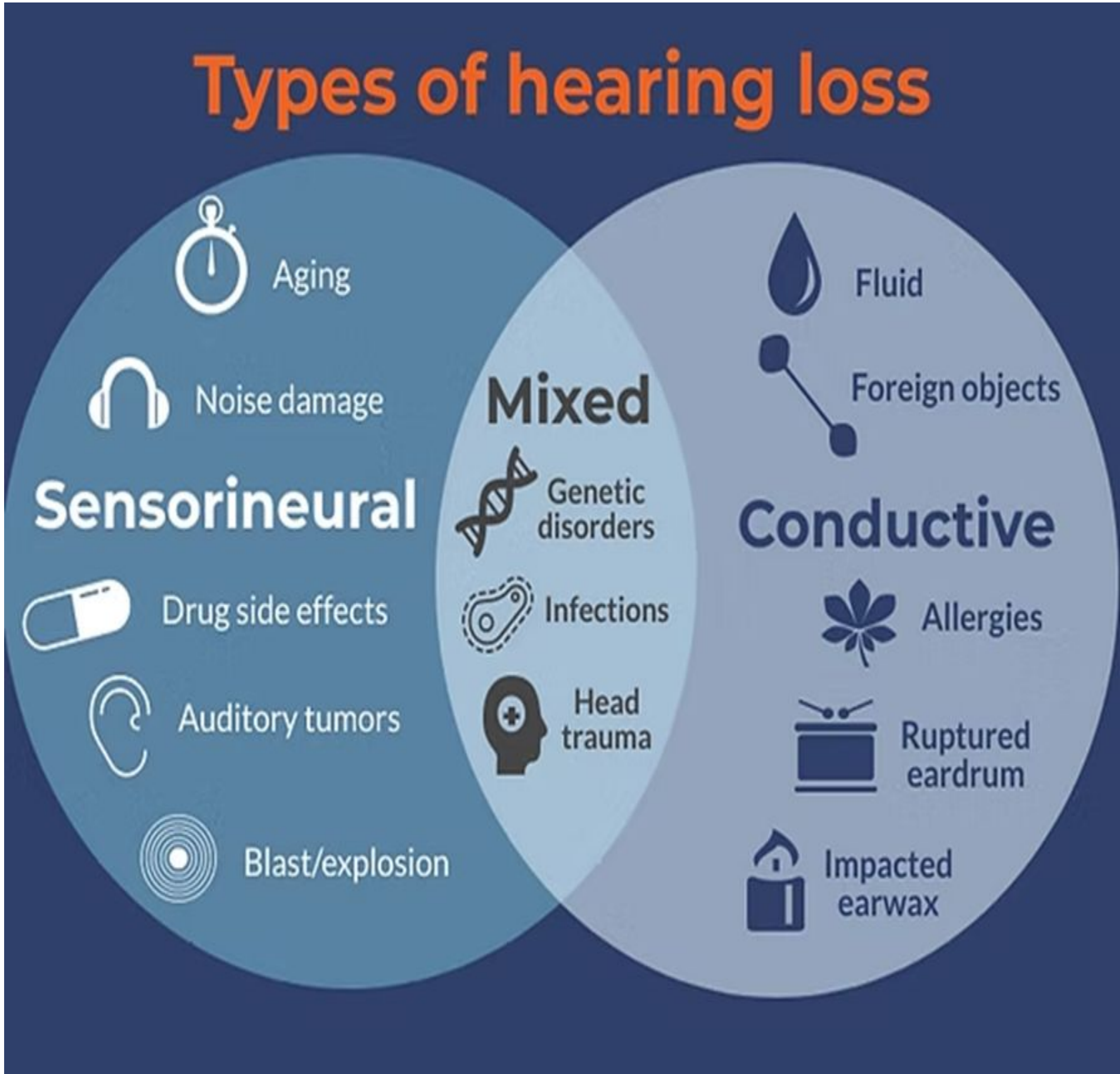
Hearing loss affects people of all ages and can be caused by many different factors.

The three basic categories of hearing loss are

Sensorineural hearing loss,

Conductive hearing loss

and Mixed hearing loss.



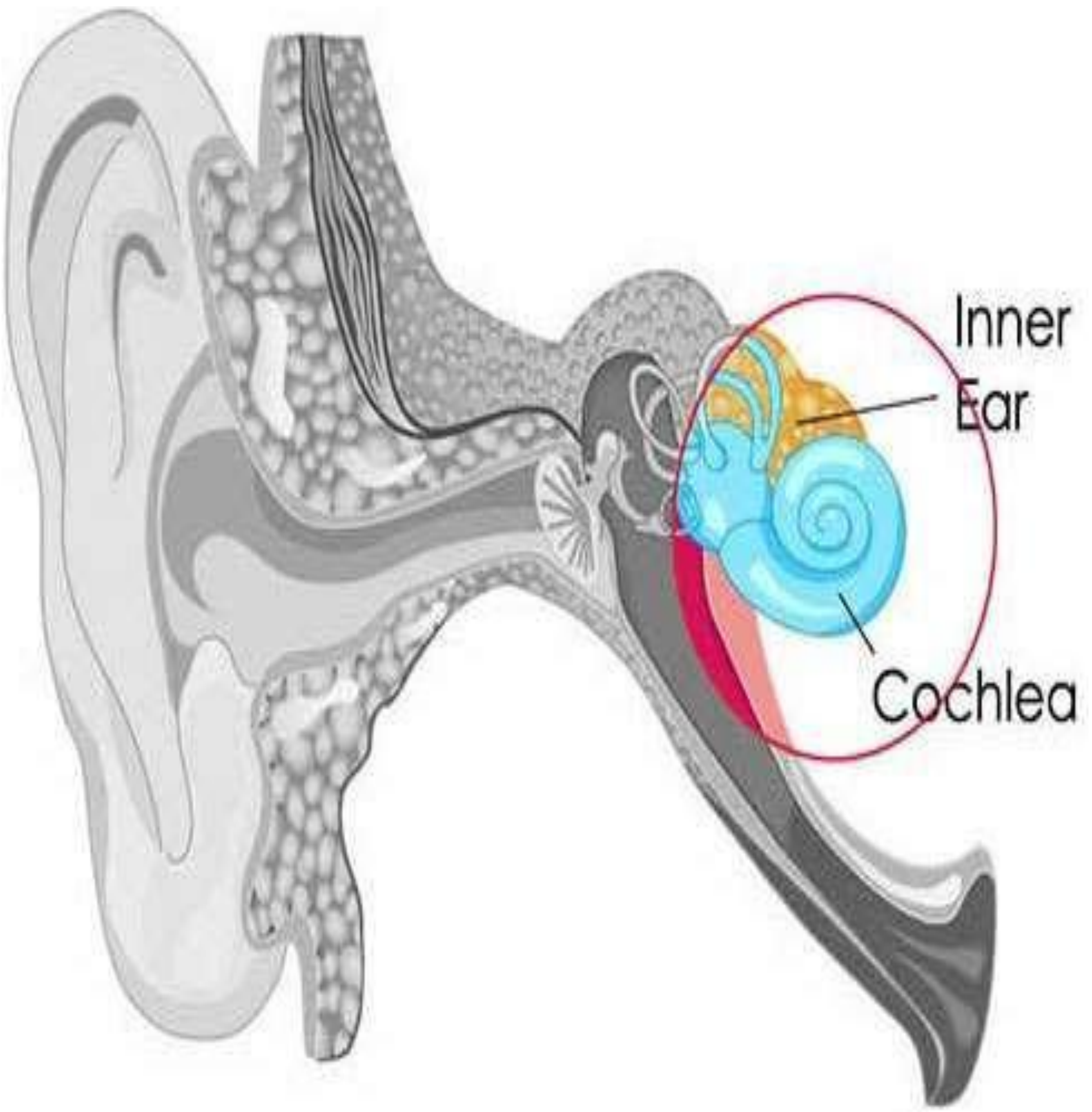
Sensorineural hearing loss

Sensorineural hearing loss is the most common type of hearing loss.

It occurs when the inner ear nerves and hair cells are damaged — perhaps due to age, noise damage or something else.

Sensory hearing loss originates in the cochlea and involves the hair cells and nerve endings.

Sensorineural hearing loss results from disease or trauma to the sensory or neural components of the inner ear.



Conductive hearing loss

Conductive hearing loss can be caused by anything that interferes with the ability of the sound wave to reach the inner ear such as ,

foreign bodies,

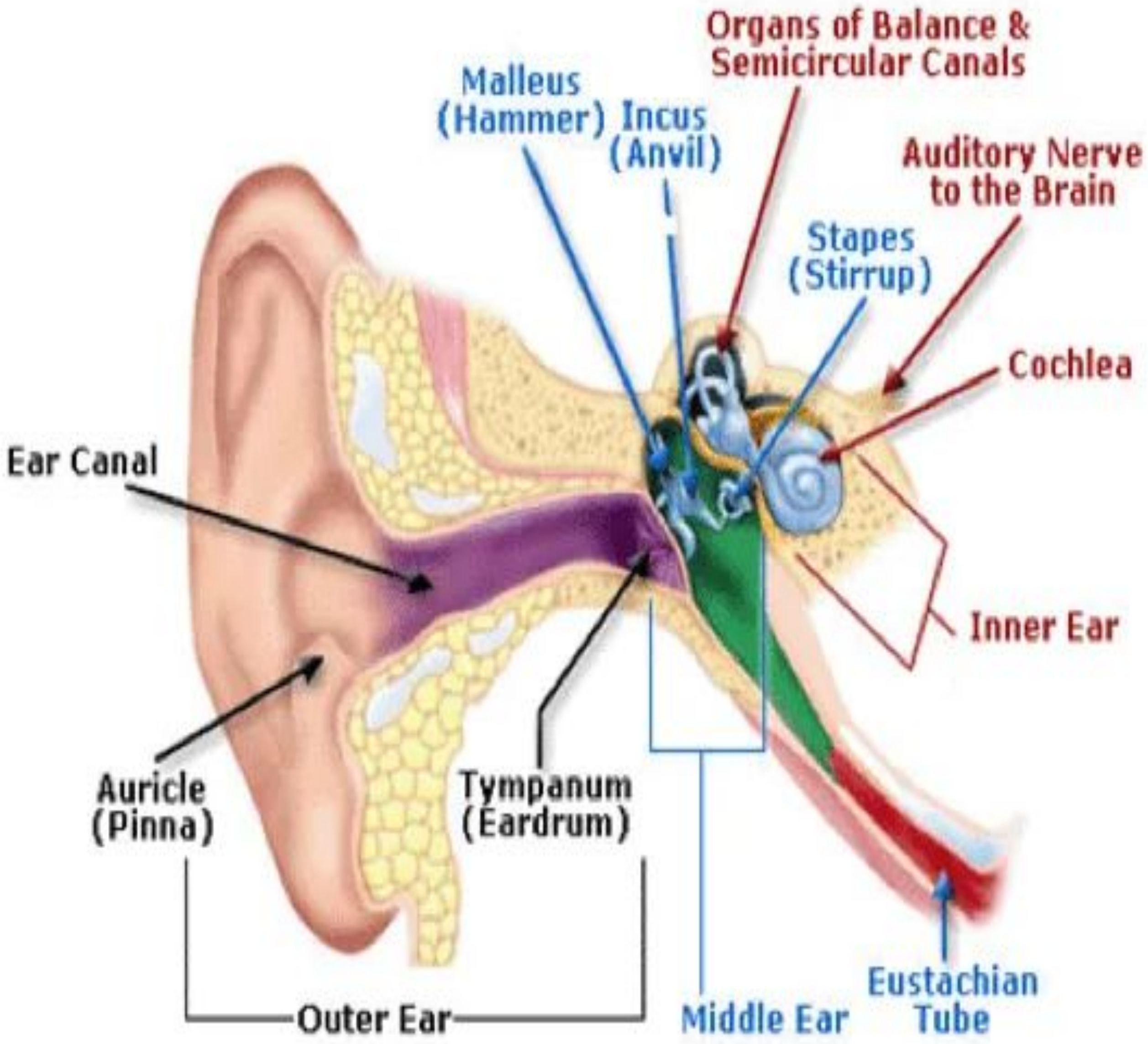
infection ,

fluid,

tumors,

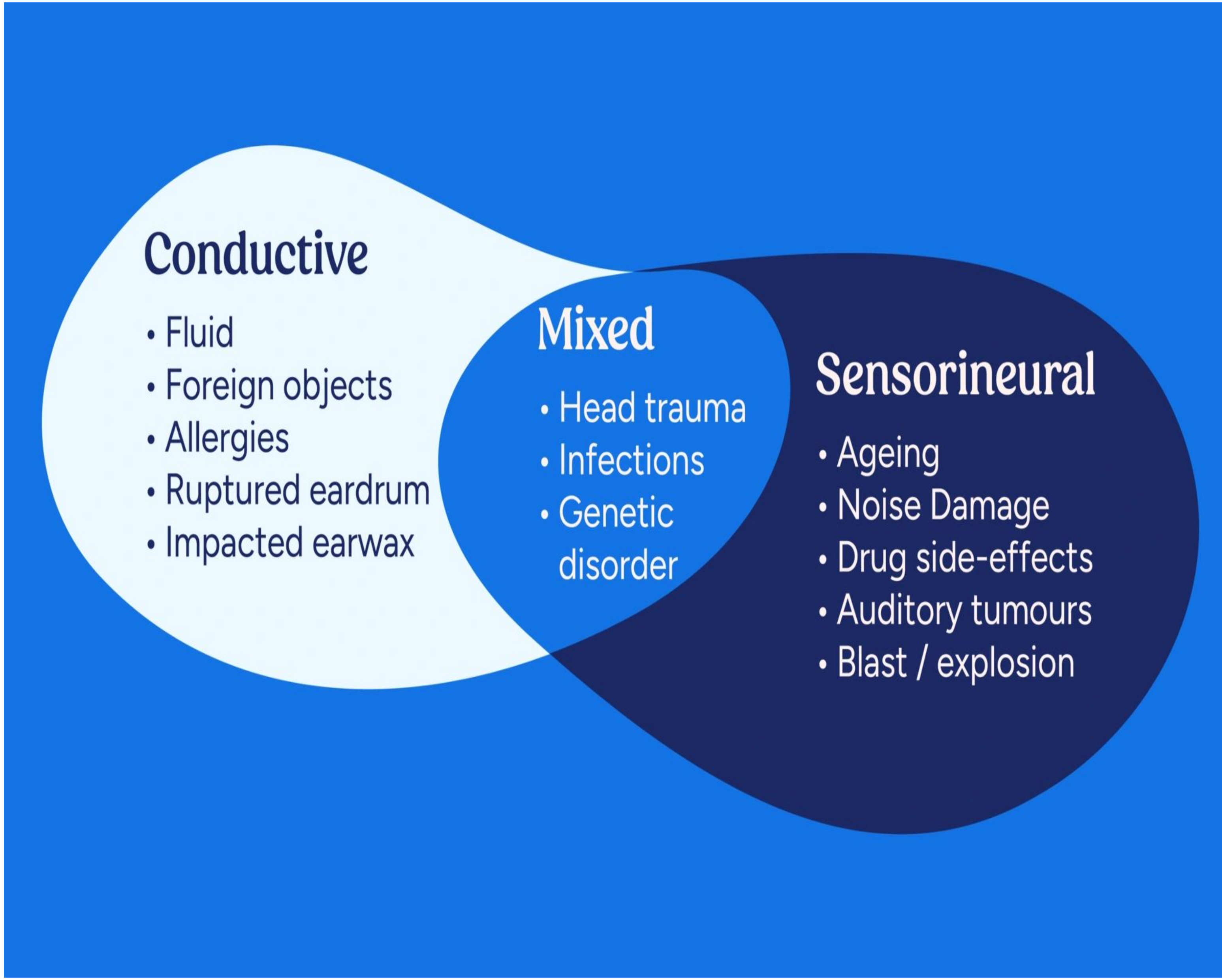
earwax

Conductive hearing loss can often be treated surgically or with medicine.



Mixed hearing loss

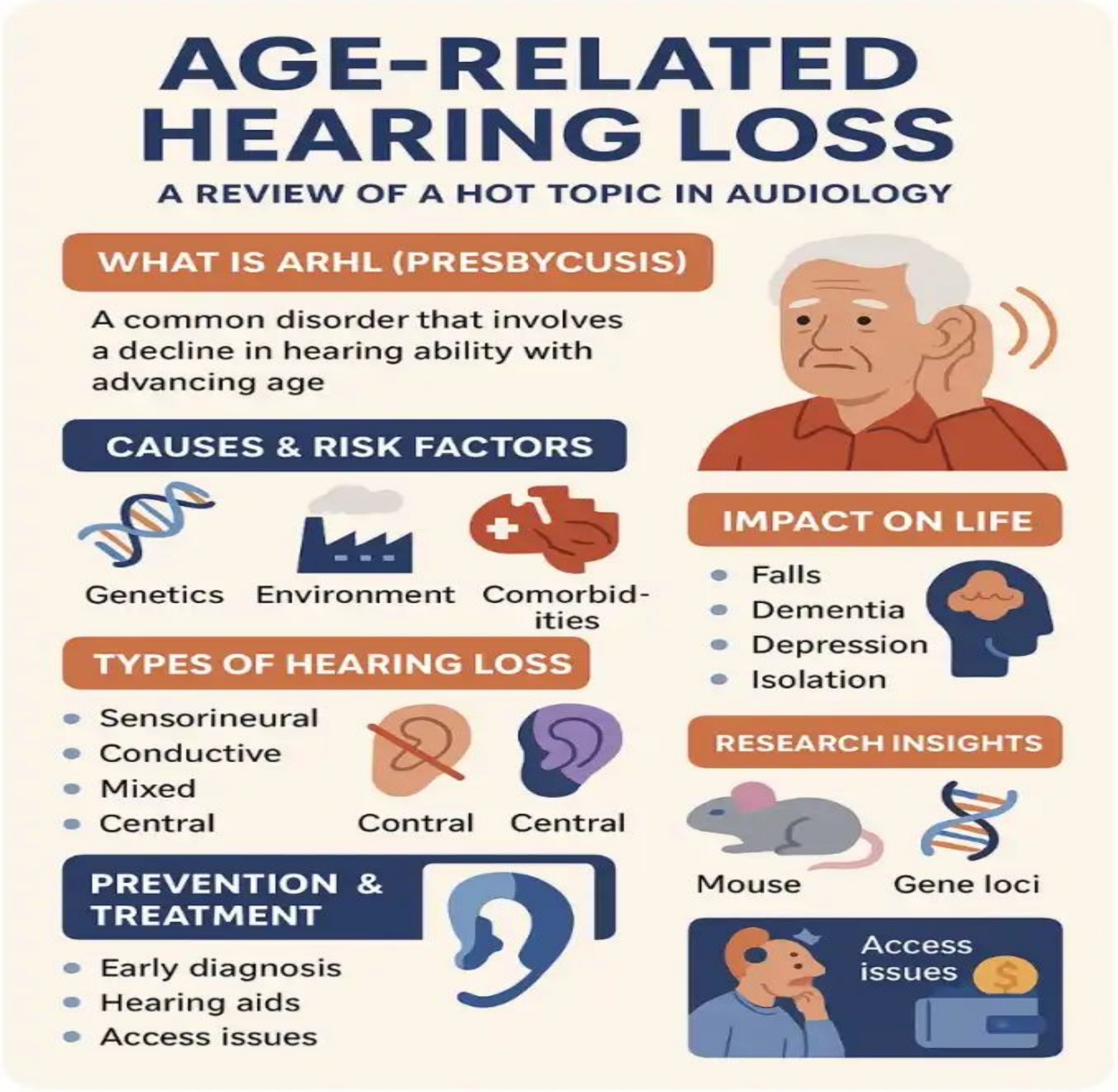
- Mixed hearing loss occurs when an individual has both conductive and sensorineural hearing loss.
- This can be caused by a combination of any of the disorders



Presbycusis

It is hearing loss caused by the aging process that results from degeneration of the organ of Corti.

This degenerative process usually begins at 50 yrs of age.



Central hearing loss

Central hearing loss occurs when the central nervous system cannot interpret normal auditory signals.

This condition occurs with such disorders as cerebrovascular accidents and tumors.

Functional hearing loss

Functional hearing loss is a hearing loss for which no organic cause or lesion can be found.

Also called psychogenic hearing loss.

Precipitated by emotional stress.

Malingering is a type of psychogenic hearing loss.

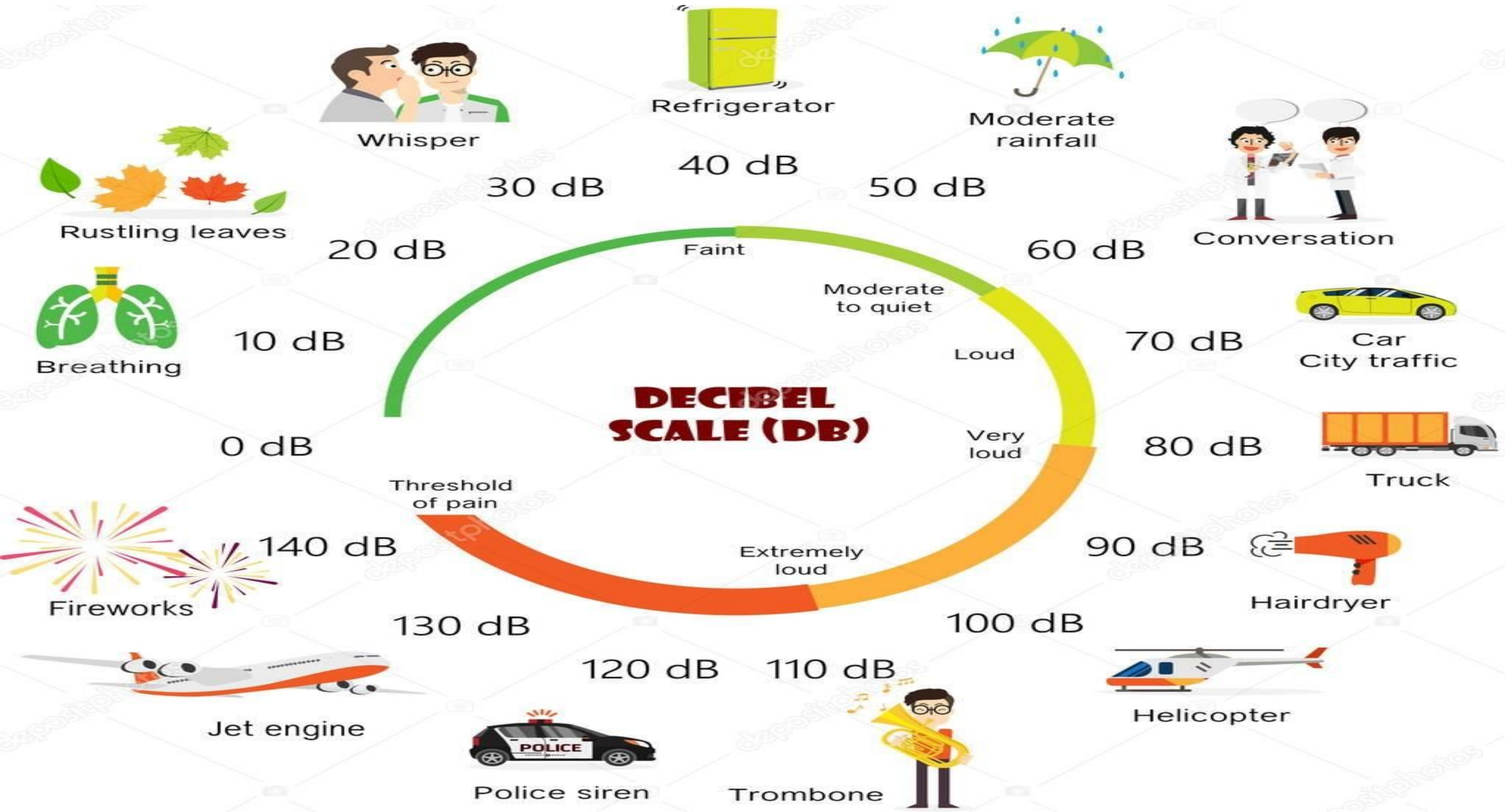
In Malingering, there is no organic or psychological cause.

The patient is pretending to be deaf for personal gains.

Severity of hearing loss

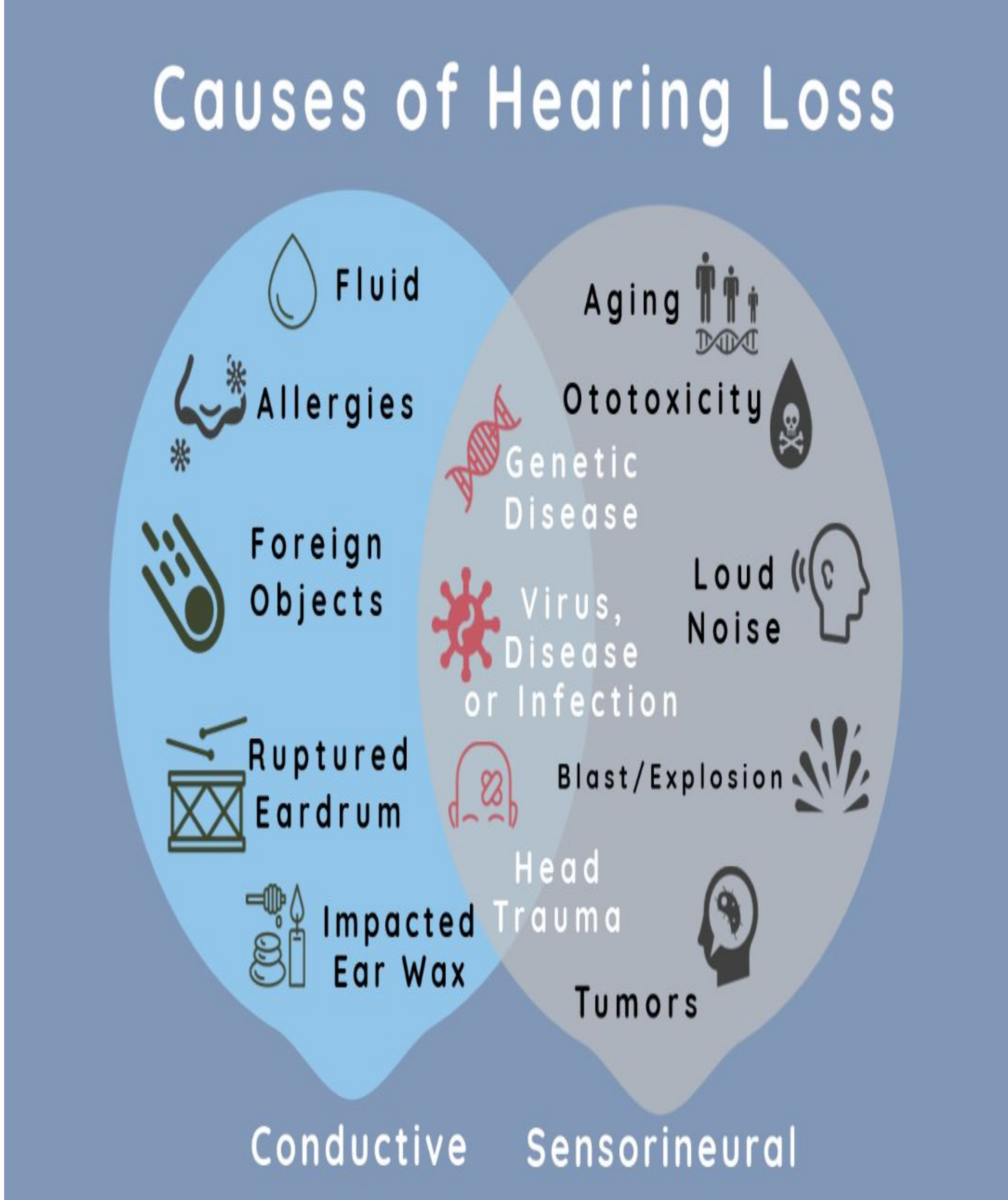
Loss in Decibels	Interpretation
0–15	Normal hearing
>15–25	Slight hearing loss
>25–40	Mild hearing loss
>40–55	Moderate hearing loss
>55–70	Moderate to severe hearing loss
>70–90	Severe hearing loss
>90	Profound hearing loss

A decibel (abbreviated **dB**) is a unit of measurement used to express the intensity or loudness of a sound


DECIBEL SCALE (DB)
Faint
Moderate to quiet
Loud
Very loud
Extremely loud
Threshold of pain
0 dB
10 dB
Breathing
20 dB
Rustling leaves
30 dB
Whisper
40 dB
Refrigerator
50 dB
Moderate rainfall
60 dB
Conversation
70 dB
Car City traffic
80 dB
Truck
90 dB
Hairdryer
100 dB
Helicopter
110 dB
Trombone
120 dB
Police siren
130 dB
Jet engine
140 dB
Fireworks
POLICE

Causes of hearing loss

- Hearing loss is caused by many factors, most frequently from natural aging or exposure to loud noise. The most common causes of hearing loss are:
- Aging
- Noise exposure
- Head trauma
- Virus or disease
- Genetics
- Ototoxicity
- Inherited Premature birthCertain conditions during birth like lack of oxygen to breath
- Conditions during pregnancysuch as rubella, syphilis, andother infections
- Certain drugs can cause deafness
- Excessive loud music Old age, to name a few



Things that can cause conductive hearing loss are:

- Infections of the ear canal or middle ear resulting in fluid or pus buildup
- Perforation or scarring of the eardrum
- Wax buildup
- Dislocation of the middle ear bones (ossicles)
- Foreign object in the ear canal
- Otosclerosis (an abnormal bone growth in the middle ear)
- Abnormal growths or tumors

Things that can cause sensorineural hearing loss are:

- Aging
- Injury
- Excessive noise exposure
- Viral infections (such as measles or mumps)
- Ototoxic drugs (medications that damage hearing)
- Meningitis
- Diabetes
- Stroke
- High fever or elevated body temperature
- Meniere's disease (a disorder of the inner ear that can affect hearing and balance)
- Acoustic tumors
- Heredity

Risk Factors

- Family history of sensorineural impairment
- Congenital malformations of the cranial structure (ear)
- Low birth weight (1500 g)
- Use of ototoxic medications (eg:gentamycin, loop diuretics)
- Recurrent ear infections
- Bacterial meningitis
- Chronic exposure to loud noises
- Perforation of the tympanic membrane

Six Signs of Hearing Loss

About **one-third of older adults** have hearing loss, and the chance of developing it increases with age.

Trouble understanding people over the phone

Finding it hard to follow conversations when two or more people are talking

Often asking people to repeat themselves

Needing to turn up the TV volume higher than other people do

Trouble understanding others because of background noise

Thinking that others seem to mumble

To learn more about hearing loss, visit www.nia.nih.gov/hearing-loss.

NIH National Institute on Aging

Causes of Conductive deafness

EXTERNAL EAR:

- Impacted Wax
- Otitis Externa
- Foreign Bodies
- Polyps
- Tumours
- Fluid in the ear

MIDDLE EAR

- Congenital defects of the ear drum and ossicles.
- Perforation of the tympanic membrane
- Dilocation of middle ear bone
- Traumatic: Barotrauma, rupture of ear drum, skull fracture
- Inflammation: AOM, COM
- Neoplasms
- Otosclerosis (an abnormal bone growth in the middle ear)

Causes of sensorineural deafness

- Presbycusis is the medical term for age-related hearing loss
- Tumours: Acoustic neuroma
- Meniere's disease
- Ototoxic drugs: streptomycin, Kanamycin, neomycin, salicylates, frusemide and quinine.

INNER EAR

- Congenital
- Trauma: Head injury, surgical injury to labyrinth, loud sounds producing concussion.
- Infections: mumps, syphilis, tuberculous meningitis, enteric fever, labyrinthitis.

Causes of Central hearing loss

Central hearing loss occurs when there is dysfunction in the central auditory nervous system (the auditory nerve, brainstem, or auditory cortex) rather than the physical ear. It affects the brain's ability to interpret and process sound, causing difficulty understanding speech and localizing sounds

- Traumatic Brain Injury (TBI)
- Cerebrovascular accidents
- Brain tumors
- Neurodegenerative Disorders: Diseases like Alzheimer's and Parkinson's can impact how the brain decodes and interprets auditory signal
- Multiple sclerosis

Clinical manifestations

- Tinnitus (the perception of noise or ringing in the ears.)
- Increasing inability to hear when in a group
- Need to turn up the volume of the television
- Failure to respond or In appropriate response to oral communications
- Excessively loud speech
- Strained facial expression
- Constant need for clarification of conversation
- Social withdrawal

Diagnostic measures

- History
- Physical examination
  - Rinne’s test
  - Weber’s test
- Audiometry
- Tympanogram

Why Diseases Can Affect Hearing

Hearing relies on a delicate system of bones, nerves, and fluids in your ear. Disruptions from disease can interfere at any point.

Some diseases affect the middle ear, blocking the passage of sound. Others impact the inner ear or the auditory nerve, which transmits sound to the brain. Still others harm blood flow, reducing the oxygen your inner ear needs to function.

Damage can lead to conductive hearing loss (blockage of sound), sensorineural hearing loss (damage to inner structures), or a mix of both. The earlier you spot the signs, the more hearing you may be able to preserve.

Which common diseases can cause hearing loss

Disease #1: Otosclerosis

Otosclerosis affects the tiny bones in the middle ear, especially the stapes. It causes abnormal bone growth that reduces the ear's ability to conduct sound. This creates a type of conductive hearing loss.

How It Leads to Hearing Loss

When the stapes cannot move freely, sound vibrations cannot reach the inner ear. This dampens sound clarity and volume.

Treatment Options

Hearing aids to amplify sound

Stapedotomy surgery to replace the stiff bone with a prosthetic

Otosclerosis can be progressive.

Early treatment helps preserve communication and quality of life.

Disease #2: Ménière’s Disease

Ménière’s disease affects the inner ear and causes fluctuating hearing loss, vertigo, and tinnitus. It results from a buildup of fluid in the inner ear.

How It Leads to Hearing Loss

The excess fluid disrupts the inner ear’s ability to transmit sound signals properly. Over time, the damage may become permanent.

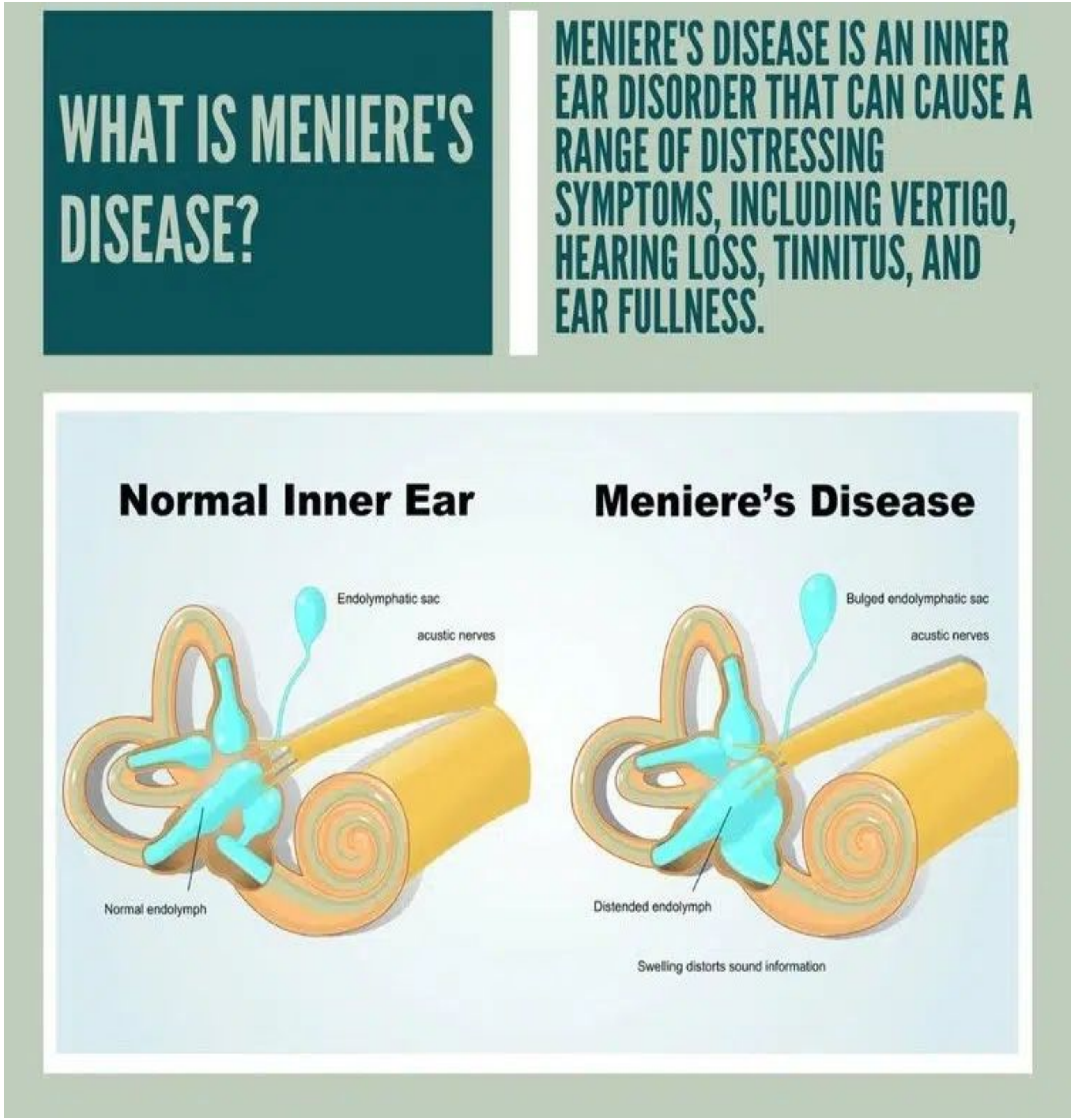
Treatment Options

Low-sodium diet and diuretics to reduce fluid

Steroid injections

Hearing aids for lasting hearing loss

Surgery for severe cases



Disease #3: Diabetes

Diabetes damages small blood vessels throughout the body, including those in the ear.

According to research published in Nature’s *Communications Medicine*, diabetes is significantly associated with hearing loss due to microvascular damage.

How Diabetes Affects Hearing

Reduces blood flow to the cochlea

Damages the auditory nerve

Increases inflammation in ear tissues

What You Can Do

Manage blood sugar levels

Have your hearing tested regularly

Treat early signs of hearing loss with hearing aids

People with diabetes are twice as likely to have hearing loss.

Early intervention makes a real difference.

Disease #4: Autoimmune Inner Ear Disease (AIED)

AIED is a rare condition where your body's immune system attacks your inner ear. It causes progressive hearing loss and often affects both ears.

How It Affects Hearing

The immune system attacks inner ear cells, damaging the hair cells responsible for sending sound to your brain.

Diagnosis and Treatment

A hearing care provider may order blood tests, MRI, and hearing evaluations.

Treatments include:

Steroids to reduce inflammation

Immunosuppressant drugs

Hearing aids to address permanent loss

AIED can cause permanent damage within weeks. If you lose hearing suddenly, don't wait. Seek care right away.

Disease #5: Acoustic Neuroma

An acoustic neuroma is a slow-growing, non-cancerous tumor on the auditory nerve. While benign, it can cause serious hearing problems if left untreated.

How It Affects Hearing

As the tumor grows, it presses on the auditory nerve. This blocks or distorts sound signals on their way to the brain.

Treatment Options

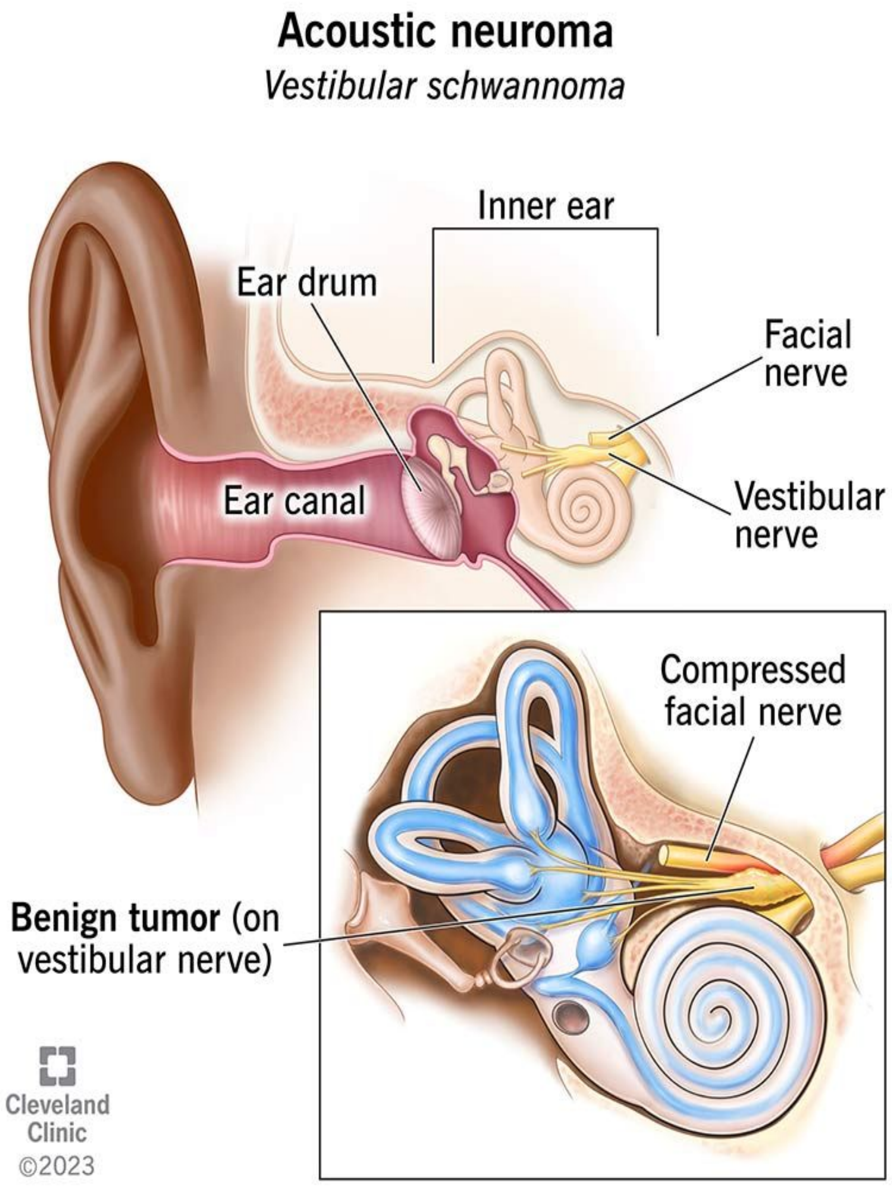
MRI to confirm diagnosis

Monitoring for small tumors

Surgical removal

Radiation therapy

Even after treatment, hearing loss may persist. That's why early detection is vital.



Disease #6: Viral and Bacterial Infections

Many infections can cause temporary or permanent hearing loss. Some damage the middle ear. Others attack the cochlea, auditory nerve, or surrounding structures.

Infections That Can Cause Hearing Loss

Otitis Media (Middle Ear Infection): Often causes temporary conductive loss

Meningitis: Can destroy inner ear structures

Measles and Mumps: Cause nerve damage in children

Cytomegalovirus (CMV): Common cause of congenital hearing loss

What to Do If You Have a Disease That May Cause Hearing Loss

Hearing loss related to disease often progresses. But with proactive care, you can protect your hearing and improve communication.

Step 1: Get a Hearing Test

Step 2: Treat the Underlying Condition

Step 3: Address Permanent Hearing Loss

Even if hearing doesn’t return, you have options.

Hearing Aids: Amplify sounds and improve speech clarity

Cochlear Implants: For profound loss

How to Prevent Disease-Related Hearing Loss

Manage Chronic Health Conditions

Control diabetes and high blood pressure

Avoid smoking

Eat a balanced diet

Stay physically active

Protect Against Infections

Keep vaccinations up to date

Treat colds and flu early


EAR NOSE & THROAT
ASSOCIATES OF LUBBOCK
4 Common Causes of
Sudden Hearing Loss in One Ear
01. AN INFECTION
02. VASCULAR OR BLOOD FLOW PROBLEMS
03. NEUROLOGICAL DISORDERS
04. A TUMOR

How to Prevent Disease-Related Hearing Loss

- Manage Chronic Health Conditions
- Control diabetes and high blood pressure
- Avoid smoking
- Eat a balanced diet
- Stay physically active
- Protect Against Infections
- Keep vaccinations up to date
- Treat colds and flu early
- Avoid exposure to people with contagious illnesses

Prevention

- Minimize the exposure to trauma,
- infection,
- ototoxic drugs.
- Avoid the risk factors.
- Wear ear protection to prevent noise-induced hearing loss when exposed to loud noise.

Management

Medical management

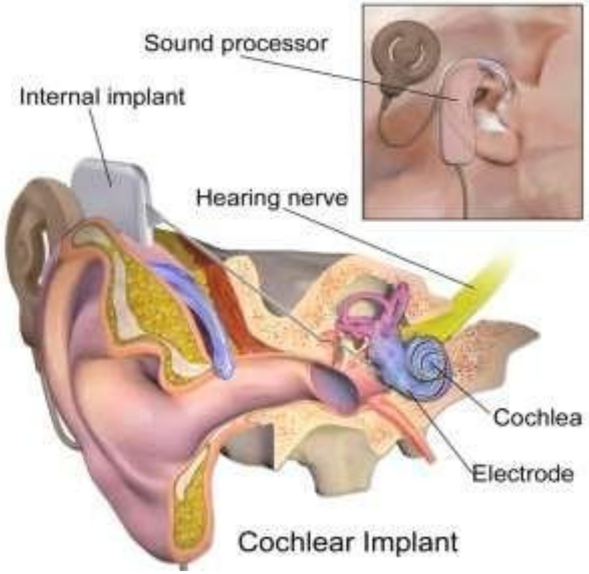
- Restore hearing
  - Antibiotics: to treat infections
  - Remove impacted wax or foreign bodies
  - Ceruminolytics for impacted wax
  - Corticosteroids for inflammation
  - Treat underlying disorders

- Assist hearing
  - Hearing aids
  - Implantable middle ear hearing devices
  - Cochlear implants
  - Sign language
  - Auditory rehabilitation

- **Hearing aids**: Hearing aid is designed to amplify sound

- **Implantable middle ear hearing devices**: Implantable middle ear hearing aids are implanted surgically, can improve sound perception for patients with moderate-to-severe sensorineural hearing loss.

- **Cochlear implants:** Cochlear implants are surgically placed electrical devices that receive sound and transmit the resulting electrical signal to electrodes implanted in the cochlear of the ear.



- **Sign language**: It involves hand shapes, movement of hands, arms, body and facial expressions

Pathophysiology

Disorders of external or middle ear

↓

Interruption in the transmission of sound by air to the inner ear

↓

Conductive deafness

↓

Disorders of inner ear or vestibulocochlear nerve or aging

↓

Interruption in neural transmission of sound to brain

↓

Sensorineural deafness